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# Devops B2

# Lab Exercise 20- Deploy a Docker Application using AWS ECR and ECS

### 1. Objective

In this lab, you will learn how to:

- Build a container image using **Docker**
- Store the container image in **Amazon Elastic Container Registry**
- Deploy the container on **Amazon Elastic Container Service**
- Access the running application through a public endpoint

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### 2. Prerequisites

Before starting the lab, ensure you have:

- AWS account with appropriate permissions
- Docker installed
- AWS Command Line Interface installed and configured
- Basic understanding of containers

- Internet access

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## 3. Lab Scenario

A development team has built a small web application using containers. Your task as a DevOps engineer is to:

1. Containerize the application using **Docker**
2. Store the container image in **Amazon ECR**
3. Deploy the container to **Amazon ECS**
4. Verify the application is running successfully

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## 4. Lab Tasks

### Task 1: Create a Docker Application

Create a simple application.

Example

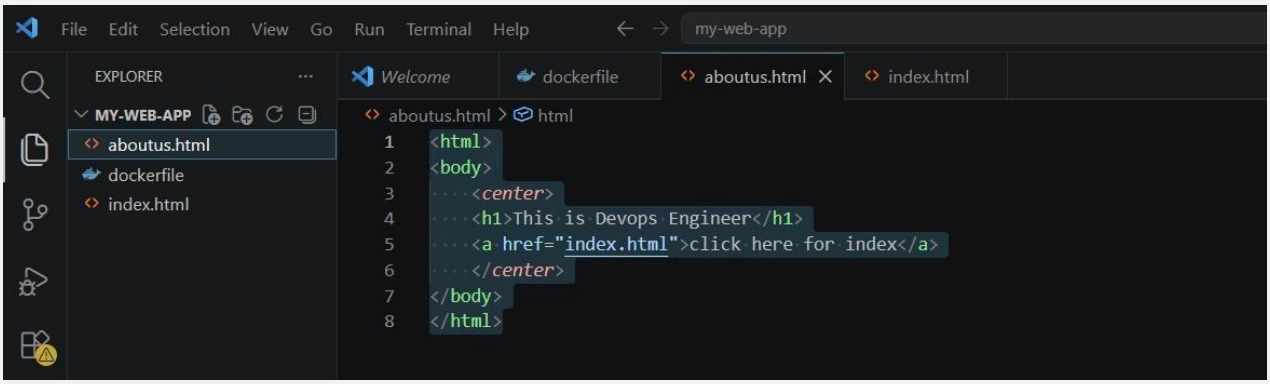
**index.html**

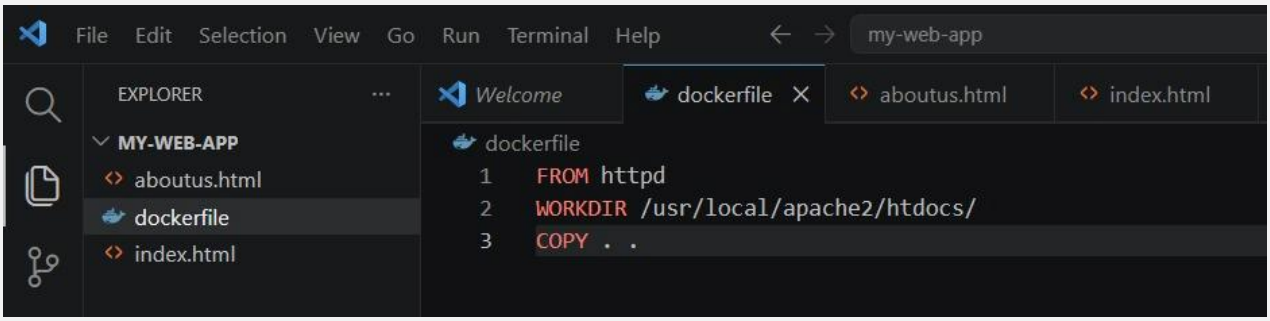
```
Hello from AWS ECS Lab
```

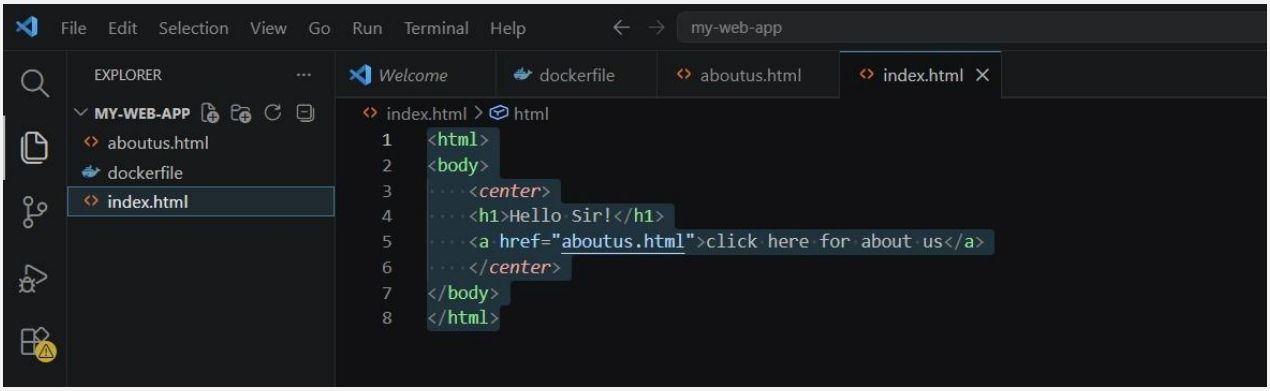
Create a **Dockerfile**

FROM nginx

COPY index.html /usr/share/nginx/html/index.html



### Task 2: Build the Docker Image

Build the image locally using **Docker**.

```
Docker build -t ecs-lab-app .
```

Verify the image.

```
docker images
```

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### Task 3: Create an ECR Repository

Create a repository in **Amazon ECR**.

Steps:

1. Open AWS Console
2. Go to **Amazon ECR**
3. Click **Create Repository**
4. Repository name:

```
ecs-lab-repository
```

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### Task 4: Authenticate Docker to ECR

Login to the ECR registry using **AWS CLI**.

Example:

```
aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin <account-id>.dkr.ecr.us-east-1.amazonaws.com
```

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### Task 5: Tag the Docker Image

Tag the image to match the ECR repository.

Example:

```
docker tag ecs-lab-app:latest <account-id>.dkr.ecr.us-east-1.amazonaws.com/ecs-lab-repository:latest
```

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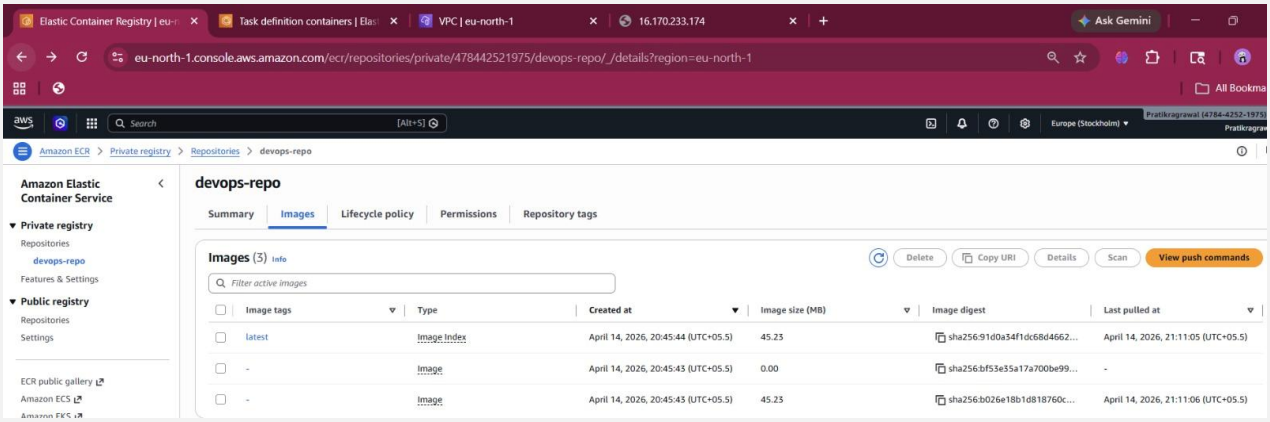
### Task 6: Push the Image to ECR

Push the image to the repository.

```
docker push <account-id>.dkr.ecr.us-east-1.amazonaws.com/ecs-lab-repository:latest
```



Verify the image in the **ECR console**.

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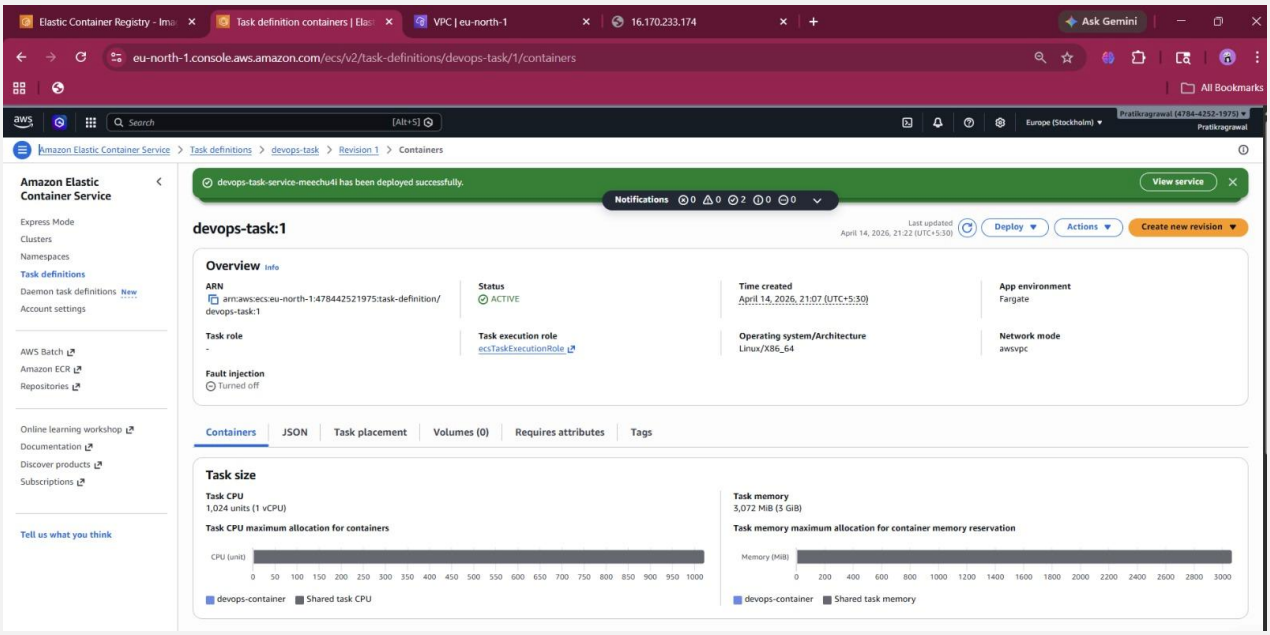
## Task 7: Create ECS Cluster

Navigate to **Amazon ECS**.

Steps:

1. Click **Create Cluster**
2. Select **Fargate**
3. Provide cluster name:

ecs-lab-cluster

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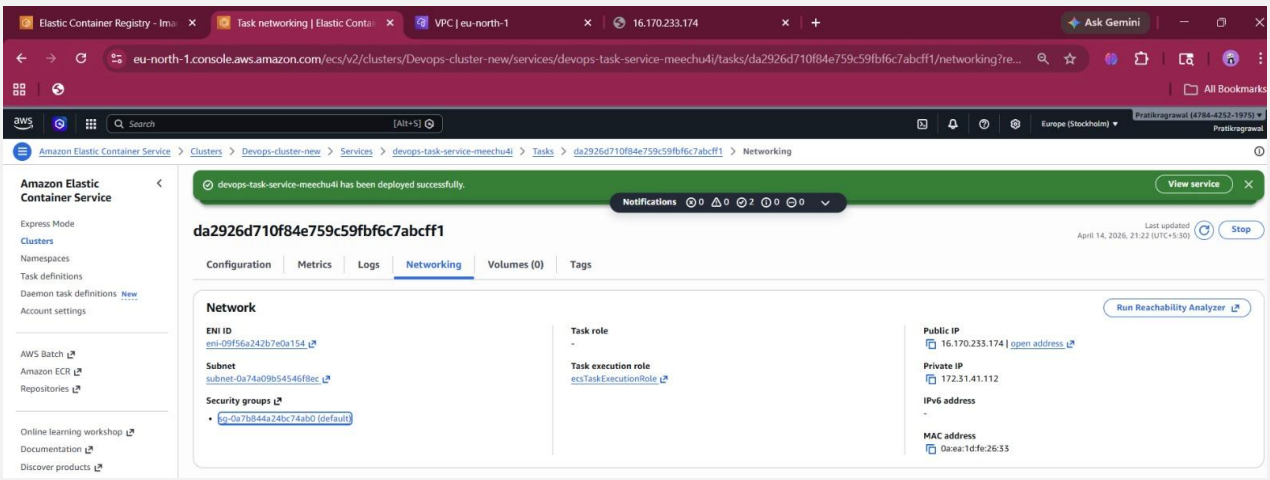
## Task 8: Create Task Definition

Create a task definition.

Settings:

Container image:

<account-id>.dkr.ecr.us-east-1.amazonaws.com/ecs-lab-repository:latest




Port mapping:

80

## Task 9: Run ECS Service

Deploy the container.

Steps:

1. Create service
2. Select **Fargate**
3. Desired tasks:
4. Enable public IP

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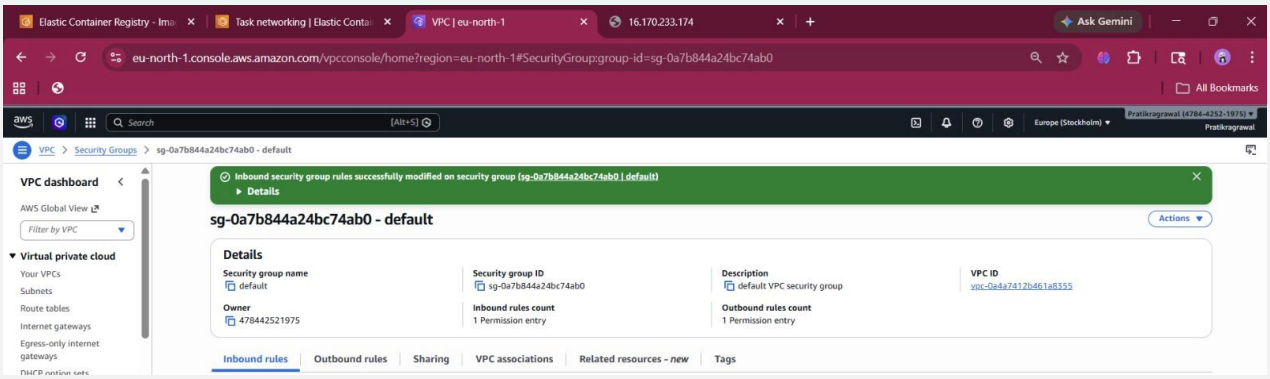
## Task 10: Verify Deployment

After deployment:

1. Open ECS service
2. Copy the **public IP or load balancer URL**
3. Open in browser

Expected output:

Hello from AWS ECS Lab

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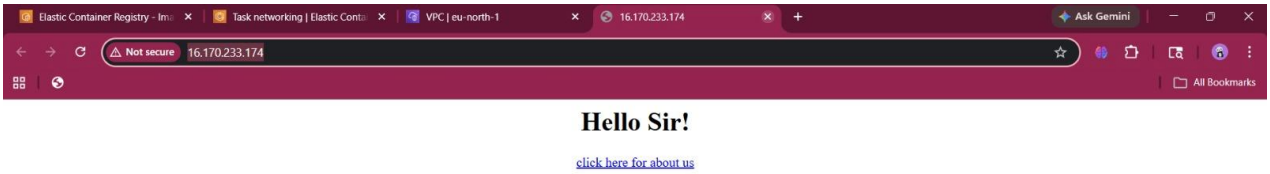
## 5. Expected Outcome

After completing this lab:

- Docker image is stored in **Amazon ECR**
- Container is running on **Amazon ECS**
- Application is accessible through a browser